

**Lemma.** Let  $G$  be a group and  $A$  be a set.

For an Action of  $G$  on  $A$ ,  $|G \cdot G_a| = |G_a|$

**Proof.** Since for any  $g_1, g_2 \in G_a$ ,  $g_1 g_2^{-1} a = g_1 a = a$ , so  $g_1 g_2^{-1} \in G_a$ .

Thus  $G_a$  is a subgroup. And, define a map

$$G_a \rightarrow \{gG_a \mid g \in G\}; g \cdot a \mapsto gG_a$$

Then,  $g_1 G_a = g_2 G_a$  implies  $g_1^{-1} g_2 \in G_a$ , or  $g_1^{-1} g_2 a = a$ , or  $g_1 a = g_2 a$ .  
So injective, and the surjective is trivial. Thus, Proved.

**Thm.** Let  $G$  be a group and  $H$  be a subgroup of  $G$ .

An action  $G \times \{gH \mid g \in G\} \rightarrow \{gH \mid g \in G\}; (g_1, g_2 H) \mapsto g_1 g_2 H$   
satisfies:

1) It acts transitively. (It is clear, since  $\{gH \mid g \in G\}$  is a partition of  $G$ ).

2)  $G_H = H$

3)  $\ker \pi_H = \bigcap_{x \in G} xHx^{-1}$  is the largest normal subgroup of  $G$  contained in  $H$ .

**Proof.**  $\ker \pi_H = \{g \in G \mid gxH = xH \text{ for all } x \in G\}$   
 $= \{g \in G \mid x^{-1}gxH = H \text{ for all } x \in G\} = \bigcap_{x \in G} xHx^{-1}$ .

Let  $g \in G$  and  $h \in \ker \pi_H$ . Then, for any  $x \in G$ ,

$$x^{-1}(ghg^{-1})x = (g^{-1}x)^{-1} \cdot h \cdot (g^{-1}x) \in \ker \pi_H, \text{ so } \ker \pi_H \trianglelefteq G.$$

And, given  $g \in \ker \pi_H$  and  $h \in H$ ,  $h^{-1}ghH = H \Rightarrow gh \in H$ , so  $\ker \pi_H \leq H$ .

Finally, if  $N \leq H$  and  $N \trianglelefteq G$ , then given  $x \in G$

$$N = xNx^{-1} \leq xHx^{-1}, \text{ so } N \subseteq \bigcap_{x \in G} xHx^{-1}$$

**Corollary.** If  $G$  is a finite group of order  $n$  and  $p$  is the smallest prime dividing  $n$ , then any subgroup of index  $p$  is normal.

**Proof.** Let  $H \leq G$  such that  $|G:H| = p$ . And, let  $K = \ker \pi_H$ ,  $\pi_H$  is an action above.

Since  $K \leq H$ , let  $|H:K| = k$ . Then,  $|G:K| = |G:H| \cdot |H:K| = p \cdot k$ .

Since  $H$  has  $p$  left cosets,  $G/K \cong \pi_H[G] \leq S_p$ , so  $pk = |G/K|$  divides  $p!$ ,

so  $k = 1$ ,  $H = K$ .

Automorphism of Group.

Prop. Let  $H$  be a normal subgroup of the group  $G$ .

Then,  $G$  acts by conjugation on  $H$  as automorphisms of  $H$ .

More specifically, for each  $g \in G$ ,

$$H \rightarrow H: h \mapsto ghg^{-1}$$

is an automorphism of  $H$ . Moreover, the permutation represent,

$$\tau_H: G \rightarrow S_H: g \mapsto \sigma_g \quad \text{where } \sigma_g(h) = ghg^{-1}$$

is a homomorphism of  $G$  into  $\text{Aut}(H) \leq S_H$ , with  $\ker \tau_H = C_G(H)$ .

In summary,  $G/\ker \tau_H = G/C_G(H) \cong \text{Im } \tau_H \leq \text{Aut}(H)$ , and  $G/Z(G) \cong \text{Inn}(G)$ .

Proof. Since  $H$  is normal, the action is well-defined. And,

$$\ker \tau_H = \{g \in G \mid \sigma_g = \text{id}\} = \{g \in G \mid ghg^{-1} = h \text{ for all } h \in H\} = C_G(H).$$

Corollary. For any subgroup  $H$  of  $G$ ,

$N_G(H)/C_G(H)$  is isomorphic to a subgroup of  $\text{Aut}(H)$ .

Prop.  $\text{Aut}(\mathbb{Z}_n) \cong \mathbb{Z}_n^\times$ .

Proof. Denote  $\mathbb{Z}_n = \langle x \rangle$ , where  $x \in \mathbb{Z}_n$ . Given automorphism  $\varphi \in \text{Aut}(\mathbb{Z}_n)$ , the  $\varphi$  is uniquely determined by  $\varphi(x) = x^a$ , the integer  $1 \leq a \leq n$ , and  $|x| = |\varphi(x)| = |x^a|$ , so  $\gcd(a, n) = 1$ . Now,

$$\text{Aut}(\mathbb{Z}_n) \rightarrow \mathbb{Z}_n^\times; \varphi_a \mapsto a \pmod n$$

is the isomorphism.

$$\mathbb{Z}_8^\times = \{1, 3, 5, 7\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Note: If  $p$  is a group of a prime order  $p$ , then

$$\text{Aut}(p) \cong \mathbb{Z}_p^\times \cong \mathbb{Z}_{p-1}.$$

**Lemma.** Let  $P \in \text{Syl}_p(G)$ . If  $Q$  is  $p$ -subgroup of  $G$ , then  $Q \cap N_G(P) = Q \cap P$ .

**Proof.** Since  $P \leq N_G(P)$ ,  $Q \cap P \leq Q \cap N_G(P)$  is trivial.

Since  $Q \cap N_G(P) \leq N_G(P)$ , denote  $H = Q \cap N_G(P)$ ,

$$PH = \bigcup_{h \in H} Ph = \bigcup_{h \in H} hP = HP$$

Thus  $PH \leq G$  and  $|PH| = \frac{|P| \cdot |H|}{|P \cap H|}$ . By the Lagrange's Thm,  $H \leq Q$  means

order of  $H$  is power of  $p$ . Therefore,  $P \leq PH$  and  $|PH|$  is power of  $p$ , thus  $P \cap H = P$ , or  $H \leq P$ . Now,  $H = Q \cap N_G(P) \leq Q$ ,  $H \leq Q \cap P$ .  $\square$

**Lemma.** If  $G$  is finite abelian group and  $p$  is a prime number dividing  $|G|$ , then  $G$  has an element of order of  $p$ .

**Proof.** Using induction for  $|G|$ . If  $|G|=1$ , trivial.

Now, suppose that the statement hold for all groups whose order is smaller than  $|G|$ .

Since  $|G|>1$ , take  $x \in G$  with  $x \neq 1$ . That is,  $|x|>1$ .

If  $|G|=p$ , then  $|x|=p$ , by the Lagrange's Theorem ( $|x| \mid p$ ).

Now, suppose  $|G| \neq p$  ( $|G|>p$ ).

If  $p$  divides  $|x|$ ,  $|x|=p^n$  for some  $n \in \mathbb{N}$ , so  $|x^p|=p$ . So clear.

If  $p$  does not divide  $|x|$ , let  $N = \langle x \rangle$ . (Clearly,  $p \nmid |N| = |x|$ ).

Since  $G$  is abelian,  $N$  is Normal, and  $x \neq 1$ , so it is non-trivial.

By the Lagrange Theorem,  $|G| = |G/N| \cdot |N|$ . But,  $p$  does not divide  $|N|$ ,

$p$  divides  $|G/N|$ . Now, using the hypothesis of the induction

$G/N$  has an element of order  $p$ , denote this  $\bar{y} = yN$ . Since  $\bar{y} \neq \bar{1}$ ,  $y \notin N$ .

But,  $\bar{y}^p = \bar{1}$ , so  $y^p \in N$ . This implies  $\langle y^p \rangle \neq \langle y \rangle$ . Since  $|y^p|$  divides  $|y|=p$ , the only possibilities are  $|y^p|=1$ , this  $y$  is we desired.

Theorem.

Let  $G$  be a group of order  $p^\alpha \cdot m$  where  $p^\alpha$  is a power of prime  $p$ , and  $p \nmid m$ .

1st sylow theorem. Sylow  $p$ -subgroup of  $G$  exists.

2nd sylow theorem. If  $P \in \text{Syl}_p(G)$  and  $Q$  is  $p$ -subgroup of  $G$ ,  
then  $Q \leq gPg^{-1}$  for some  $g \in G$ .

3rd sylow theorem.  $n_p \equiv 1 \pmod p$  and  $n_p = |G : N_G(P)|$  for any  $P \in \text{Syl}_p(G)$ .

Proof. To show the Existence of Sylow  $p$ -subgroup, using induction for  $|G|$ .

If  $|G|=1$ , then  $p^\alpha=1$ , so  $\text{Syl}_p(G) = \{G\} \neq \emptyset$ .

Assume that the sylow  $p$ -subgroup always exists for any group of order smaller than  $G$ .

If order of  $Z(G)$  is divided by  $p$ , then by the Cauchy's Theorem for abelian group, there exists order of  $p$  element  $y \in Z(G)$ . Let  $N = \langle y \rangle$ . Denote  $\overline{G} = G/N$ .

By the hypothesis and the Lagrange's Theorem,  $|\overline{G}| = |G|/|N| = p^{\alpha-1} \cdot m$ .

By the induction,  $\overline{G}$  has a sylow  $p$ -subgroup  $\overline{P}$  of order  $p^{\alpha-1}$ .

By the correspondence isomorphism Theorem, there is  $P \leq G$  such that  $\overline{P} = P/N$ .

By the Lagrange's Theorem again,  $p^{\alpha-1} = |\overline{P}| = |P/N| = |P|/p$ , so  $|P| = p^\alpha$ .

Therefore, there exists a sylow  $p$ -subgroup  $P$ , in case of  $p \mid |Z(G)|$ .

If order of  $Z(G)$  is not divided by  $p$ , let  $g_1, \dots, g_r$  be representatives of distinct non-central conjugacy classes of  $G$ . The class equation gives:

$$|G| = |Z(G)| + \sum_{i=1}^r |G : C_G(g_i)|$$

If all  $|G : C_G(g_i)|$  are divided by  $p$ , then  $|Z(G)|$  must be divided by  $p$ , contradiction.

Therefore, for some  $1 \leq i \leq r$ ,  $p$  does not divide  $|G : C_G(g_i)|$ . Let  $H = C_G(g_i)$ .

By the Lagrange's theorem,  $|H|$  divides  $|G|$ , so  $|H| = p^\alpha \cdot k$  where  $p \nmid k$ ,  $k \mid m$ .

(If  $|H|$  does not have factor of  $p^\alpha$ , then  $p$  divides  $|G : C_G(g_i)|$ , which is contradiction.)

And, since  $g_i \notin Z(G)$ ,  $H < G$ . Therefore, by the assumption of induction,

$H$  has a sylow  $p$ -subgroup, so it is sylow  $p$ -subgroup of  $G$ .

To show the second and third assertion, for group  $G$  of order  $p^a m$ , there exists a Sylow  $p$ -subgroup  $P$  of  $G$ , by the first assertion. Let

$$S = \{gPg^{-1} \mid g \in G\} = \{P_1, P_2, \dots, P_r\}$$

and let  $Q$  be any  $p$ -subgroup of  $G$ .

Let  $Q$  acts by conjugation on  $S$ . Write  $S$  as a disjoint union of orbit:

$$S = \mathcal{O}_1 \cup \dots \cup \mathcal{O}_s \quad \text{where } |\mathcal{O}_1| + \dots + |\mathcal{O}_s| = r$$

(Note that  $S$  has only one orbit under  $G$ , but it may have more than one under  $Q$ )

Reorder the elements of  $S$  so that the first  $s$  numbers elements of  $S$  are

representatives of the  $Q$ -orbit  $\mathcal{O}_i$ :  $P_i \in \mathcal{O}_i$ .  $(P_1, P_2, \dots, P_s, P_{s+1}, \dots, P_r)$   
 It follows that  $|\mathcal{O}_i| = |Q : N_Q(P_i)|$ .  
 $(\underbrace{P_1, P_2, \dots, P_s}_{\mathcal{O}_1}, \underbrace{P_{s+1}, \dots, P_r}_{\mathcal{O}_s})$

By definition,  $N_Q(P_i) = N_G(P_i) \cap Q$  and  
 by the Lemma above,  $= P_i \cap Q$ .

Therefore,  $|\mathcal{O}_i| = |Q : P_i \cap Q|$ ,  $1 \leq i \leq s$ . Since  $Q$  was arbitrary, take  $Q = P_1$ . Then  
 $|\mathcal{O}_1| = |Q : Q \cap Q| = 1$ . Now, for  $2 \leq i \leq s$ ,  $P_1 \cap P_i < P_1$ , so

$$|\mathcal{O}_i| = |P_1 : P_1 \cap P_i| > 1 \quad \text{for all } 2 \leq i \leq s.$$

Since  $P_1$  has order of power of  $p$ ,  $|\mathcal{O}_i|$  must be power of  $p$ . Therefore,

$$r = \underbrace{|\mathcal{O}_1|}_{1} + \underbrace{|\mathcal{O}_2| + \dots + |\mathcal{O}_s|}_{\text{order of } p} \equiv 1 \pmod{p}. \quad \dots (*)$$

Let  $Q$  be any  $p$ -subgroup of  $G$ . Suppose that  $Q$  is not contained in  $P_i$  for all  $i$ .

Then,  $Q \cap P_i < Q$  for all  $1 \leq i \leq r$ , therefore  $|\mathcal{O}_i| = |Q : Q \cap P_i| > 1$  for each  $1 \leq i \leq s$ .

Since  $Q$  has order of power of  $p$ , so  $|\mathcal{O}_i| > 1$  is power of  $p$ ,

therefore all  $|\mathcal{O}_i|$  ( $1 \leq i \leq s$ ) are divided by  $p$ , it contradicts to  $(*)$ .

Consequently,  $Q \leq gPg^{-1}$  for some  $g \in G$ , which proves the second.

Let  $Q$  be any Sylow  $p$ -subgroup of  $G$ . By the argument above,  $Q \leq gPg^{-1}$

for some  $g \in G$ . And since  $|Q| = |gPg^{-1}|$ ,  $Q = gPg^{-1}$ .

Therefore, all Sylow  $p$ -subgroups are conjugate. Finally, for any  $p \in \text{Syl}_p(G)$ ,

$$\# \text{Syl}_p(G) \stackrel{\text{def}}{=} n_p = |G : N_G(p)|, \quad \text{since } \text{Syl}_p(G) \text{ is the orbit.}$$

Moreover,  $m = |G : p| = |G : N_G(p)| \cdot |N_G(p) : p| = n_p \cdot |N_G(p) : p|$ , so  $n_p \mid m$ .  $\square$

Corollary.

Let  $P$  be a Sylow  $p$ -subgroup of  $G$ . Then the following are equivalent.

1).  $P$  is the unique Sylow  $p$ -subgroup of  $G$ .

2).  $P$  is normal in  $G$ .

3).  $P$  is characteristic in  $G$ .

Proof. Start 1). Since  $\text{Syl}_p(G) = \{P\}$ , the Sylow theorem gives that for all  $g \in G$ ,  $gPg^{-1} \in \text{Syl}_p(G)$ , that is,  $gPg^{-1} = P$ . Therefore  $P \trianglelefteq G$ .

Conversely, given  $Q \in \text{Syl}_p(G)$ ,  $Q$  is conjugate of  $P$ , that is,  $Q = gPg^{-1}$  for some  $g \in G$ . Since  $P \trianglelefteq G$ ,  $Q = gPg^{-1} = P$ .

And, since the automorphism preserves the order, therefore  $1) \Rightarrow 3)$ .

Finally,  $3) \Rightarrow 2)$  trivially, by the inner automorphism, so  $3) \Rightarrow 1)$ .

Group of order of 15 is cyclic.

Let  $G$  be a group of order 15.

By the Sylow theorem, there are only one Sylow 3-subgroup and Sylow 5-subgroup, denote  $P, Q$  respectively. ( $n_3 \equiv 1 \pmod{3}$  and  $n_3 | 5 \Rightarrow n_3 = 1$ )

Let  $xy \in G$  be generators of  $P, Q$  respectively.

Since  $P \trianglelefteq G$  is normal,  $G = N_G(P)$

Meanwhile, consider an action of  $G$  on  $P$  as

$$G \times P \rightarrow P : (g, h) \mapsto ghg^{-1}.$$

And the afforded representation  $\varphi: G \rightarrow S_P$ . Then,  $\ker \varphi = C_G(P)$ , therefore  $G/\ker \varphi = G/C_G(P)$  is a subgroup of  $\text{Aut}(P)$ . But, since  $P \cong \mathbb{Z}_3$ ,  $\text{Aut}(P)$  has order of 2, therefore  $G/C_G(P)$  must be trivial, that is,  $C_G(P) = G$ .  
Now,  $x$  and  $y$  commute, so  $|xy| = 15$ , so  $G = \langle xy \rangle$ .

To show  $G$  must be abelian, suppose that  $Z(G) = \{1\}$ .

By the Lagrange's theorem, every element of  $G$  has order 3 or 5, except 1.

If all non-identity elements have order 3, then the class equation gives

$$15 = 1 + \sum |G : C_G(g_i)| = 1 + k \cdot 5 \text{ for some } k \in \mathbb{N},$$

because  $C_G(g_i)$  must contains  $\langle g_i \rangle \cong \mathbb{Z}_3$  and if  $C_G(g_i) > \langle g_i \rangle$ , then  $C_G(g_i) = G$  by the Lagrange's theorem, it contradicts to  $g_i \notin Z(G)$ .  
But,  $15 - 1 = 14$  is divided by 5 but 1 is not divided, therefore,  $G$  must have order 5 element. But this  $x \in G$ .

[Group of order of  $pg$  ( $p < g$  are prime,  $p \nmid g-1$ ).

If  $Z(G) \neq 1$ , then  $G/Z(G) \cong \mathbb{Z}_p$  or  $\mathbb{Z}_g$  by the Lagrange Theorem, then  $G$  must be abelian, since  $G/Z(G)$  is cyclic.

Now, suppose that  $Z(G) = 1$ .

By the Lagrange Thm, every non-identity element of  $G$  has order  $p$  or  $g$ .

If there is no order  $g$  element, then the class equation gives

$$pg = 1 + \sum [G : C_G(g_i)] = 1 + kg,$$

because  $C_G(g_i) = \langle g_i \rangle \cong \mathbb{Z}_p$ . (Since  $g_i \notin Z(G)$ ,  $C_G(g_i) \neq \langle g_i \rangle$ ).

This is contradiction, since  $g \nmid 1$ . So,  $G$  has a order  $g$  element, denote this to  $x$ .

Let  $H = \langle x \rangle$ . Since  $[G : H] = p$  is the smallest prime dividing  $|G|$ , so  $H$  is normal.

Now,  $G/H = N_G(H)/C_G(H)$  is a subgroup of order  $p$ , and isomorphic to a subgroup of  $\text{Aut}(H)$ .

Since  $\text{Aut}(H) \cong \mathbb{Z}_g^\times$  has order of  $g-1$ , this implies  $p \nmid g-1$ , contradiction.

[Group of order of  $pg$ ,  $p < g$ . ( $p, g$  are prime)

Let  $P \in \text{Syl}_p(G)$  and  $Q \in \text{Syl}_g(G)$ .

Since  $n_g$  divides  $P$ ,  $n_g = 1$  or  $P$ , but  $n_g \equiv 1 \pmod{g}$ , thus  $n_g = 1$ .  $\therefore Q \trianglelefteq G$ .

1. Case of  $P \nmid g-1$ .

By the Sylow Thm,  $n_p = 1$  or  $g$ , but if  $n_p = g$ , then  $g \equiv 1 \pmod{p}$ ,  $p \mid g-1$ , contradiction.

Therefore,  $n_p = 1$ ,  $P \trianglelefteq G$ . Now,  $G$  must be a cyclic, because:

Since  $|P|$  and  $|Q|$  are primes, both are cyclic. Let  $P = \langle x \rangle$  and  $Q = \langle y \rangle$ .

Since  $P \trianglelefteq G$ ,  $N_G(P) = G$ . Therefore,

$$G/C_G(P) = N_G(P)/C_G(P) \cong S \text{ for some } S \leq \text{Aut}(\mathbb{Z}_p)$$

But,  $|\text{Aut}(\mathbb{Z}_p)| = p-1$ , thus  $S$  is trivial. (because  $|S|$  divides  $|G| = pg$ ).

Now,  $G = C_G(P)$ , and  $x \in P$ ,  $x$  and  $y$  are commute. This implies  $|xy| = pg$ .  $G = \langle pg \rangle$ .

2. Case of  $P \mid g-1$

Let  $Q$  be a Sylow  $g$ -subgroup of  $S_g$ , the symmetric group of letters.

Since  $g$  is prime,  $|N_{S_g}(Q)| = g \cdot (g-1)$ , because

1). Since  $|S_g| = g!$ ,  $|Q| = g$ .

2).  $S_g$  has  $(g-1)!$   $g$ -cycles.

3). Since order of  $gQg^{-1}$  is  $g$ , these are all cyclic.

4).  $gQg^{-1} \cap Q = \{1\}$  for all  $g \in G$ , so

$$n_g = \# \{gQg^{-1} \mid g \in G\} = (g-1)! / (g-1) = (g-2)!$$

$$\text{Therefore } n_g = G_Q = [S_g : N_G(Q)] = (g-2)!, \text{ or } N_{S_g}(Q) = g(g-1). \quad \overset{p}{\uparrow}$$

Now, since  $p \mid g-1$ ,  $p$  divides order of  $N_{S_g}(Q)$ , so it has a subgroup of order  $p$ .

And, since  $p \leq N_{S_g}(Q)$ ,  $pQ$  is subgroup of order  $pg$ .

And,  $C_{S_g}(Q) = Q$ ,  $pQ$  is non-abelian group.

$$|G : N_{S_g}(Q)| = G_Q$$

Group of order 48 is not simple.

$48 = 2^4 \cdot 3$ . Therefore,

$$n_3 = 1 \text{ or } 4 \text{ or } 16 \quad \text{and} \quad n_2 = 1 \text{ or } 3.$$

If  $n_2 = 1$ , then  $\text{PG Syl}_2(G)$  is the Normal, so there is nothing to prove.

Suppose  $n_2 \neq 1$ , or  $n_2 = 3$ . That is, there are three Sylow 2-subgroup,

$$P_1, P_2, P_3 \in G. \quad (\#P_i = 16).$$

For each  $1 \leq i < j \leq 3$ ,

$$|P_i P_j| = \frac{|P_i| \cdot |P_j|}{|P_i \cap P_j|} = \frac{2^8}{|P_i \cap P_j|} \leq 48 = |G| \quad \text{if } i, j, P_i = P_j$$

So,  $\frac{2^4}{3} \leq |P_i \cap P_j|$ ,  $|P_i \cap P_j|$  cannot be 1, 2, 4, 16 therefore

$$|P_i \cap P_j| = 8.$$

Take  $Q = P_1 \cap P_2$ , as subgroup of order 8. Since  $Q \subseteq P_1, P_2$  and index 2,  $Q \trianglelefteq P_1$  and  $Q \trianglelefteq P_2$ . That is, for any  $g \in P_1 \cup P_2$ ,

$$gQg^{-1} = Q.$$

therefore,  $P_1 \cup P_2 \subseteq N_G(Q)$ . Since  $|P_1 \cap P_2| = 8$ ,

$$|P_1 \cup P_2| = |P_1| + |P_2| - |P_1 \cap P_2| = 24$$

Now, if  $|N_G(Q)| = 24$ , then it has index 2 to  $G$ ,

$$N_G(Q) \trianglelefteq G$$

And, if  $|N_G(Q)| = 48$ , then  $N_G(Q) = G$ , so  $Q \trianglelefteq G$ .

[Group of order 30 is not simple.

$30 = 2 \cdot 3 \cdot 5$ , let  $P \in \text{Syl}_5(G)$  and  $Q \in \text{Syl}_3(G)$ .

If either  $P$  or  $Q$  is normal, then  $PQ \leq G$ , and it is group of order 15, because  $|PQ| = \frac{|P| \cdot |Q|}{|P \cap Q|} = 15$ . (All elements of  $P$  has order 5 except identity, and "  $Q$  " 3 " 1 ).

So,  $PQ$  is index of 2 to  $G$ , so  $PQ$  is normal subgroup.

Now, assume neither  $P$  nor  $Q$  is normal. By the Sylow Theorem,

$n_5 = 6$  and  $n_3 = 10$ . By the Lagrange's Theorem, for any  $P_1, P_2 \in \text{Syl}_5(G)$ ,

$P_1 \cap P_2 = \{1\}$ . So, there are  $6 \cdot 4 = 24$  distinct order 5 elements.

Similarly, there are 20 distinct order 3 elements,  $24 + 20 = 44 \leq 30$ ,

Contradiction. Therefore, either  $P$  or  $Q$  must be normal, so  $PQ$  is index 15 subgroup.

In sum. Group of order 30 must have a subgroup of order of 15.

[Group of order 105 is Not simple.

$105 = 3 \cdot 5 \cdot 7$ , so

$$n_3 = 1 \text{ or } 7$$

$$n_5 = 1 \text{ or } 21$$

$$n_7 = 1 \text{ or } 15$$

If it is simple, then  $n_3 = 7$ ,  $n_5 = 21$ ,  $n_7 = 15$ . Therefore there are

$$\underbrace{2 \cdot 7}_{\text{order 3}} + \underbrace{4 \cdot 21}_{\text{order 5}} + \underbrace{6 \cdot 15}_{\text{order 7}} = 14 + 84 + 90 \geq 130$$

So, Contradiction by Counting

[Group of order 12 is Not Simple.

Every group of order 12 has a normal sylow 3-subgroup or  $G \cong A_4$ .

$12 = 2^2 \cdot 3$ . If  $n_3 = 1$ , then  $G$  is not simple, it has a normal sylow 3-subgroup.

Suppose that  $n_3 \neq 1$ . Then, there is only possibility is  $n_3 = 4$ .

Let  $P \in \text{Syl}_3(G)$ . By the sylow Theorem,  $|G/N_G(P)| = n_3 = 4$ ,

so  $P = N_G(P)$ , by  $P \leq N_G(P)$  and  $|N_G(P)| = 12/4 = 3$ , given by the Lagrange Thm.

Now, consider an action of  $G$  on  $\text{Syl}_3(G)$  by conjugation:

$$G \times \text{Syl}_3(G) \rightarrow \text{Syl}_3(G) : (g, P) \mapsto gPg^{-1}$$

Since  $\text{Syl}_3(G) = \{P_1, P_2, P_3, P_4\}$ , we can write the permutation representation of this action:

$$\mathcal{Q}: G \rightarrow S_4 : g \mapsto \sigma_g \quad \text{where} \quad \sigma_g: S_4 \rightarrow S_4 : P_i \mapsto gP_i g^{-1}$$

Now, the elements of  $\ker \mathcal{Q}$  normalizes all sylow 3-subgroups. That is,

$$\ker \mathcal{Q} \leq N_G(P) = P$$

Since  $P$  is not normal in  $G$ ,  $\ker \mathcal{Q} = 1$ . That is,  $\mathcal{Q}$  is injective,

$$G \cong \mathcal{Q}[G] \leq S_4.$$

Since  $G$  contains 8 elements of order 3, by counting,

and  $S_4$  has exactly 8 element of order 3, and all these are even,

Thus  $G \cong A_4$ .

[Groups of  $p^2 \cdot q$ ,  $p$  and  $q$  distinct primes.

Let  $G$  be a group of order  $p^2 \cdot q$ . Let  $P \in \text{Syl}_p(G)$  and  $Q \in \text{Syl}_q(G)$ .

[In case of  $p > q$ , then  $n_p = 1$  or  $1+kp$  ( $k \geq 1$ ), but  $1+kp \nmid q$ , so  $n_p = 1$ .

Thus  $P \trianglelefteq G$ .

□ If  $|G|=60$  and  $G$  has more than one Sylow 5-subgroup, then  $G$  is simple.

Proof.  $60 = 5!$ . Since  $n_5 > 1$ , then only possibility is  $n_5 = 6$ .

Assume there is non-trivial proper normal  $H \triangleleft G$  exists.

If  $5 \mid |H|$ , then it has at least one Sylow-5 subgroup, and since  $H \triangleleft G$ , all Sylow 5-subgroups are contained in  $H$ , so,

$$|H| \geq 1 + 4 \cdot 6 = 25,$$

therefore  $|H| = 30$ . but, since order of 30 has a unique normal Sylow 5-subgroup, so contradiction. Thus,  $5 \nmid |H|$ .

If  $|H| = 6$  or  $12$ ,